

TED UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

SENIOR PROJECT

PROJECT SPECIFICATIONS REPORT

ERDENIS: *Reinforcement Learning Based Attack and Evasion Decision-Making for UAVs*

https://erdenis.com.tr

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1. Introduction

1.1 Description

1.1.1 Problem Statement and Motivation

As unmanned aerial vehicles (UAVs) become increasingly deployed in contested airspaces, they will need to be capable of responding to hostiles that may appear during their operation. Currently, the majority of the existing UAVs require a remote operator to respond to any encountered threat. In case of a close-range combat, a few seconds is the amount of time that is left to take a decision. Moreover, a delay in communication or a malfunction of a data link may lead to insufficient time left for the operator to react. Therefore, a UAV requires a decision-making module that would perform the task autonomously by analyzing the current tactical situation and performing an adequate maneuver: evading the enemy if under threat, attacking if at the advantage and proceeding with the mission otherwise.

Existing methods of solving the problem are based on hand-crafted rules, expert systems and pursuit-evasion control laws. All of these methods are easy to interpret yet brittle, since they incorporate a limited number of maneuvers and cannot deal with non-linear flight dynamics and the large state space of an air battle. An alternative approach involves reinforcement learning, in which maneuvering behavior is learned through interaction with a simulated environment. Recent research shows that hierarchical multi-agent reinforcement learning (HMARL) can produce viable air combat behavior through the separation of tactical actions and maneuver control [1], [2] and that RL agents can compete in realistic dogfights [3].

1.1.2 Project Objective

ERDENIS aims at developing and assessing the performance of a decision-making framework using reinforcement learning that would enable a UAV to conduct attacks and evasion maneuvers in a realistic 3D simulation environment ensuring that all performed maneuvers remain within the flight envelope of the UAV. The developed framework will be assessed in comparison to standard rule-based frameworks on metrics of success in attacking, evasion, collision avoidance and flight envelope compliance.

1.1.3 Operational Scenario

The scenario envisages a combat UAV (UCAV) conducting a mission in accordance with a planned route of way points in the 3D airspace. In the process of mission execution the UAV can encounter one or more hostile aircraft. The UAV needs to make decisions continuously about whether to continue with its mission or to evade the threat, and to attack when the geometry allows for it. Engagement scenarios that are obligatory in the project are 1-vs-1 and 2-vs-2. Larger n-vs-n engagements are optional extensions. Weapons are represented by a gun with WEZ and per step probability of hit [2].

1.1.4 Proposed Approach

ERDENIS adopts a **two-level hierarchical architecture**, inspired by [1] and [2]:

- **High level: Commander policy.** For each UAV, the commander observes the tactical situation (own state, nearest opponents and nearest friendly aircraft) and selects one option: **Navigate** (continue the mission), **Evade** or **Attack**. In the 2-vs-2 configuration, the Attack option also specifies which opponent to engage. The commander is invoked at fixed intervals and on tactical events (such as an aircraft being destroyed or entering a weapon engagement zone).
- **Low level: Maneuver policies.** Three low-level policies output continuous flight control commands (aileron, elevator, rudder, throttle) and, for the Attack policy, a fire command. The **Navigate** policy flies the aircraft toward mission waypoints, the **Evade** policy maximizes survival

against one or more threats, the **Attack** policy maneuvers into a firing position behind the opponent and engages it.

Low-level policies are trained first using **curriculum learning** with increasingly capable opponents (from stationary or randomly moving targets to scripted opponents, self-play and league-play against the mixture of already trained policies). After that, the commander is trained with the low-level policies frozen. Following the approach used in our prior work on this problem, the commander is first implemented as a **rule-based commander** using geometric advantage measures (aspect angle, antenna train angle, distance) and an assignment step for target selection in multi-aircraft engagements; it is then replaced by a **learned (RL) commander**. The rule-based commander also serves as a baseline.

Policies are trained with on-policy actor-critic methods [5], [8] under the Centralized Training and Decentralized Execution (CTDE) paradigm: each agent acts on its local observation, while a centralized critic uses global information during training, and aircraft of the same type share policy parameters. The primary training algorithm is **Multi-Agent Simple Policy Optimization (MA-SPO)** [2], [7], which outperformed MAPPO [6] in the reference study on the same problem domain; MAPPO is used as the comparison algorithm. The final choice will be confirmed empirically during development.

1.1.5 Simulation Environment and Vehicle Model

While AirSim and Gazebo were mentioned as examples of simulators in the project proposal, no particular simulator was selected. In the current specification, the selection of the simulation environment is made as JSBSim [4]. It is an open-source 6-DoF flight dynamics model and will be used as the physics engine and will be wrapped as a multi-agent environment in accordance with the Gymnasium API [9]. There are three reasons for this choice. First, JSBSim simulates fixed-wing aircraft using tabulated aerodynamic coefficients, which directly supports the requirement that maneuvers remain within the flight envelope. Second, it runs headless and much faster than real time, which is necessary because RL training requires tens of millions of environment steps [2], general-purpose simulators that are tied to a real-time rendering loop make this infeasible within the project schedule. Third, it is the simulator used in the reference works this project builds upon [2], [3], which allows for meaningful comparison. Three-dimensional visualization is provided separately by exporting flight logs in the ACMI format for replay in Tacview [15], with a real-time FlightGear connection as an optional extension.

Air-to-air attack and evasion requires a high-performance, jet-powered airframe and aerodynamic data for operational combat UAVs is not publicly available. Therefore, the UCAV flight dynamics model is derived from the validated JSBSim F-16 aerodynamic model, which is based on published NASA wind-tunnel data [11]. The model is modified to represent an unmanned vehicle: mass characteristics are adjusted considering the absence of cockpit and pilot equipment; the flight envelope of the UCAV model is limited by the structural capabilities of the airframe instead of pilot tolerance.

1.1.6 Scope and Development Phases

As required by the course, the project will be executed in two semesters:

- **Fall semester (current):** Development of the JSBSim-based multi-agent environment, the UCAV model, geometry and reward modules with unit tests, the flight envelope protection layer, scripted opponents, ACMI logging and training of the **Navigate** policy, which learns basic flight and movement skills directly from control surface commands. The semester ends with a demonstration of the trained policy flying in the 3D environment.

- **Spring semester:** curriculum training of the **Evade** and **Attack** policies, implementation of the rule-based commander, training of the learned commander, integration of the complete hierarchical system and systematic comparison against rule-based baselines.

1.1.7 Expected Deliverables

- The 3D multi-agent air engagement simulation environment based on JSBSim, containing 1-vs-1 and 2-vs-2 mission scenario configurations.
- UCAV flight dynamics model and flight envelope protection module.
- Trained low-level maneuver policies (Navigate, Evade, Attack) and the commander policy.
- Rule-based baseline implementation and comparison report.
- 3D replays of representative engagements and the project web page (https://erdenis.com.tr).

1.1.8 Summary of Specification Decisions

Topic	Decision
Simulator	JSBSim with a Gymnasium-style multi-agent wrapper
3D visualization	Tacview replay via ACMI logs, FlightGear as optional extension
Vehicle	UCAV flight model derived from the JSBSim F-16 aerodynamic model
Scenario	UCAV on a waypoint mission, engaged by hostile aircraft
Engagement sizes	1-vs-1 and 2-vs-2 mandatory, n-vs-n optional
Architecture	Two levels: commander (Navigate / Evade / Attack) and low-level maneuver policies
Action space	Continuous control surface and throttle commands, plus fire command
Commander	Rule-based first (also a baseline), then learned
Algorithm	MA-SPO as primary, MAPPO for comparison, CTDE with shared parameters
Threats	Hostile aircraft with gun WEZ, guided missiles optional
Compute	Local multi-core workstations, cloud CPU instances if needed

Table 1. Specification decisions.

1.2 Constraints

1.2.1 Economic Constraints

The project does not have a specific budget defined. All software components used within the project are free and open-source (JSBSim, Python, PyTorch, Gymnasium), and Tacview is used in its free version. Training sessions will take place using personal computers of the team members. Considering that the main cost in this problem's RL training is not related to neural network calculation but environment simulation on CPUs, multi-core CPU power is the main resource: one of the reference papers estimates that 10 to 30 hours of training is needed per curriculum level using 32-core CPU [2]. If the local resources turn out to be insufficient, the team will rent cloud CPU instances from their own budget. For keeping the cost constraints, simulation throughput will be measured before renting, preemptible (spot) instances will be favored and the training process will use frequent checkpoints for allowing interruptions.

1.2.2 Environmental Constraints

ERDENIS is a purely software project that does not include any prototype or flight testing. Simulation development replaces real flight testing, thus avoiding consumption of fuel and carbon dioxide emissions. The environmental impact will only be caused by the electricity consumption during training. This impact is reduced through profiling before large training sessions, CPU-oriented parallel simulation instead of using energy-intensive GPU clusters, re-use of trained checkpoints between curriculum levels, and avoiding duplicating experiments.

1.2.3 Social Constraints

Autonomous decision-making in military applications is a publicly sensitive topic. Thus, the project should be presented clearly and the boundaries of simulation-only project should be shown explicitly. At the same time, the project makes a contribution to the research and education in autonomous systems and reinforcement learning, and the team will document their approach to provide comprehensible and reproducible results.

1.2.4 Political Constraints

The project belongs to the defense field and therefore falls under national regulations and international restrictions on military technology. In order to follow these constraints, the project makes use of public information and open source software only. No classified and restricted information is used, and the resulting system can be considered as a research prototype that is not going to be deployed.

1.2.5 Ethical Constraints

The possibility to attack implies consideration of the issue of autonomous use of force. While the unmanned vehicle does not have a pilot on board, it is operated by a human operator from a command post. Following the ethical constraint that an autonomous system must provide for sufficient levels of human judgment over the use of force [17], ERDENIS considers the option to attack as a tactic recommendation and maneuvering capability in simulation; in any real system the decision to launch a weapon will be taken by a human operator. Integration with real weapon systems is out of scope of this project. The Evade and Navigate capabilities that affect the survival of the vehicle itself raise no such issues.

1.2.6 Health and Safety Constraints

As the project involves simulation only, there is no danger of physical harm. However, the safety is still the key design issue because any policy that is learned using reinforcement learning can take advantage of a simulation environment that wouldn't be applicable physically for certain conditions and constraints. For that reason, the project includes an explicit flight envelope protection layer (a safety shield [14]) that screens the control commands and prevents them from breaking such hard limits as minimum altitude, airspeed and load factor restrictions, in addition to limiters present in the aircraft flight control system model. Besides that, the team acknowledges that any policy verified in a simulation will require thorough verification and certification prior to its deployment on a real aircraft.

1.2.7 Manufacturability Constraints

Being a software-based project, manufacturability is defined as the ability to implement, deploy and integrate the system. The learned policies are represented by small neural networks capable of running in real time on a standard CPU. The environment is based on a standard reinforcement learning interface and supports logging in a standard format (ACMI). The main limitation is the simulation-to-reality gap: policies learned on the F-16-based aerodynamic model will need to be retrained or adapted to work on another airframe.

1.2.8 Sustainability Constraints

In order to promote maintainability and reusability outside of the scope of this project, the system is designed to be modular, including the modules of environment, aircraft, geometry, rewards, policy, and training. Configuration files will be implemented for use to set up all of the scenario and training parameters, unit tests will be created for computationally expensive modules, as well as version control and documentation. In terms of sustainability, JSBSim is maintained, while AirSim is archived [16].

1.2.9 Technical and Time Constraints

- **Time:** There are about ten weeks left until the end of the fall semester and another three months available in the spring semester. However, the spring semester timeline is tight. Therefore, the environment should be developed and verified before the end of the fall semester.
- **Compute:** Training time is limited by the number of CPU cores; the total training time for all policies will be more than 200 hours of compute. The use of rental cloud technologies could help us to stay in the designated timeframe if needed.
- **Data availability:** Aerodynamics data for real combat UAVs are not available to the public due to being classified for national security reasons; thus the vehicle model cannot be that accurate.
- **Learning stability:** Multi-agent reinforcement learning in continuous 3D space is sensitive to the reward design and curriculum.

1.3 Professional and Ethical Issues

The team commits to the ACM Code of Ethics and Professional Conduct [12], the IEEE Code of Ethics [13] and the Software Engineering Code of Ethics [18] . For ERDENIS, the most significant issues are as follows:

- **Honesty and accuracy of results** (ACM 1.3, 2.5): Results will be reported truthfully, including failed experiments and limitations. Performance claims will be based on evaluation over a large number of episodes and multiple training seeds with confidence intervals, rather than on selected favorable runs.
- **Intellectual property and attribution** (ACM 1.5): The system will be implemented by the team from scratch. Published papers and open-source repositories [1], [2], [10] are used as references for design and understanding. Any third-party component that is reused, like the JSBSim aircraft data, will be used in accordance with its license and cited. Source code without an explicit open-source license will not be copied.
- **Confidentiality** (ACM 1.7): Team members will respect confidentiality obligations from past professional and internship engagement and no proprietary or classified information will be included in the project or the publications.
- **Avoiding harm and public good** (ACM 1.1, 1.2; IEEE I.1): Given the defense context, the scope is limited to simulation, the role of human judgment in the use of force is explicitly preserved and safety mechanisms are part of the design rather than an afterthought.
- **Professional competence** (ACM 2.2, IEEE II.6): Team members will acquire the skills required for the project, follow sound engineering practices such as unit testing and code review, and seek guidance from the supervisor where the work exceeds their expertise.
- **Teamwork and shared responsibility** (SE Code, Principle 7): Work will be distributed fairly, contributions will be traceable through version control and all members share responsibility for the correctness of the final product.

- **Reproducibility:** Fixed random seeds, versioned configuration files and saved checkpoints will allow every reported result to be regenerated.

2. Requirements

Requirements are prioritized using the MoSCoW scheme: **Must** requirements are mandatory for project success, **Should** requirements are important and planned but not critical and **Could** requirements are optional extensions that will be addressed if time permits.

2.1 Functional Requirements

2.1.1 Simulation Environment

ID	Requirement	Priority
FR-1	The system shall provide a 3D multi-agent simulation environment based on the JSBSim 6-DoF flight dynamics model, exposing a Gymnasium-compatible multi-agent interface (reset, step, observations, rewards, termination).	Must
FR-2	The environment shall support 1-vs-1 and 2-vs-2 engagement configurations. Larger n-vs-n configurations through shared policies are optional extension.	Must (n-vs-n: Could)
FR-3	The environment shall generate mission scenarios consisting of a route of waypoints, with randomized initial positions, headings, altitudes and speeds for all aircraft.	Must
FR-4	The environment shall compute relative geometry between aircraft, including distance, aspect angle (AA), antenna train angle (ATA) and closure rate.	Must
FR-5	The environment shall model gun engagement through a conical Weapon Engagement Zone with configurable range, angle and per-step hit probability.	Must
FR-6	The environment shall model guided missile threats with a proportional navigation guidance law.	Could
FR-7	An episode shall terminate when all aircraft of one team are destroyed, when the mission is completed or when the time horizon is reached.	Must
FR-8	The environment shall label the outcome of each episode for every aircraft using the following definitions. (a) **Successful evasion:** the UCAV survives the engagement without being destroyed, without ground or mid-air collision and without violating hard flight envelope limits, until the threat is destroyed, the threat disengages (remains outside a configurable threat range and geometry for a configurable duration) or the time horizon ends. (b) **Successful attack:** an assigned opponent is destroyed by the UCAV's own fire while the UCAV itself survives. (c) **Mission completion:** the UCAV reaches all mission waypoints within a configurable distance tolerance and time limit, including resuming the route after an evasion or attack phase. (d) **Failure:** destroyed by opponent, ground collision, mid-air collision, boundary exit or timeout. An engagement is counted as a **win** if all opponents are destroyed and at least one UCAV survives, a **loss** if all UCAVs are destroyed and **draw** otherwise.	Must
FR-9	All scenario parameters (map size, team composition, weapon parameters, time horizon) shall be configurable through configuration files.	Must

Table 2. Simulation environment requirements.

2.1.2 Vehicle Model and Flight Envelope Protection

ID	Requirement	Priority
FR-10	The system shall provide a UCAV flight dynamics model derived from the JSBSim F-16 aerodynamic model, adapted for an unmanned configuration. The model shall be verified to achieve trimmed level flight and to perform basic maneuvers (climb, descent, coordinated turn) before policy training.	Must
FR-11	A flight envelope protection layer shall filter control commands to prevent violation of hard limits: minimum altitude above ground, minimum and maximum airspeed, and structural load factor limits (nominally +9 g / -3 g). The layer shall trigger an automatic recovery maneuver when the aircraft approaches the minimum altitude limit.	Must
FR-12	The reward functions shall penalize operation near soft envelope limits so that policies learn to remain inside the envelope without relying on the protection layer.	Should

Table 3. Vehicle model and flight envelope requirements.

2.1.3 Decision-Making Policies

ID	Requirement	Priority
FR-13	The **Navigate** policy shall fly the UCAV along mission waypoints by issuing continuous aileron, elevator, rudder and throttle commands.	Must
FR-14	The **Evade** policy shall maneuver the UCAV to avoid being destroyed by one or more threatening aircraft, using continuous control commands.	Must
FR-15	The **Attack** policy shall maneuver the UCAV into a firing position behind an assigned opponent and decide when to fire.	Must
FR-16	A rule-based commander shall select Navigate, Evade or Attack for each UCAV based on geometric advantage measures, and shall assign targets in multi-aircraft engagements.	Must
FR-17	A learned commander shall be trained with frozen low-level policies to select options and targets.	Should
FR-18	The commander shall be invoked at a fixed decision interval and additionally on tactical events (aircraft destroyed, aircraft entering a WEZ, approach to envelope limits).	Must
FR-19	Each agent shall act only on its local observation at execution time (decentralized execution).	Must

Table 4. Decision-making policy requirements.

2.1.4 Observations and Rewards

ID	Requirement	Priority
FR-20	Low-level policies shall observe the UCAV's own state (position, altitude, attitude, velocities, load factor, remaining ammunition), the relative geometry (distance, AA, ATA, closure rate) of the nearest opponents and the nearest friendly aircraft, and for the **Navigate** policy, the relative position of the next waypoint. The commander shall observe the own state, the mission progress and the relative geometry of the nearest opponents and friendly aircraft. All observations shall be normalized.	Must
FR-21	Each low-level policy shall have a task-specific reward: waypoint progress for **Navigate**, threat distance and survival for **Evade**, and positional advantage and kills for **Attack**, with additional event penalties for being destroyed, crashing or leaving the operational area. The commander shall be rewarded for mission completion, kills and survival, accumulated over the duration of each selected option.	Must

Table 5. Observation and reward requirements.

2.1.5 Training

ID	Requirement	Priority
FR-22	Policies shall be trained with MA-SPO under the CTDE paradigm with a centralized critic and parameters shared among aircraft of the same type.	Must
FR-23	MAPPO shall be implemented as a comparison algorithm using identical networks and hyperparameters where applicable.	Should
FR-24	Low-level policies shall be trained with a curriculum of increasing opponent difficulty: (L1) stationary or randomly moving targets, (L2) scripted opponents, (L3) self-play, (L4) league-play against a mixture of previously trained policies. Each level shall be initialized from the final checkpoint of the previous level.	Must
FR-25	Low-level policies shall be trained in 1-vs-1 configurations (the Navigate policy without opponents), and the commander shall be trained in 2-vs-2 configurations with the low-level policies frozen.	Must

Table 6. Training requirements.

2.1.6 Evaluation and Baselines

ID	Requirement	Priority
FR-26	The system shall implement rule-based baselines: scripted evasive maneuvers (such as break turn, beam, notch) for evasion; pure pursuit and lead pursuit for attack and the rule-based commander for decision-making.	Must
FR-27	The system shall include a non-hierarchical (flat) RL agent as an ablation baseline.	Should
FR-28	The evaluation module shall report evasion success rate, kill rate, win / draw / loss rates, collision rate (ground and mid-air), flight envelope violation rate, survival time, time-to-kill and mission completion rate, using the outcome definitions in FR-8.	Must
FR-29	Each configuration shall be evaluated over at least 1,000 episodes per scenario and over at least three independently trained seeds, and results shall be reported with 95% confidence intervals.	Must

Table 7. Evaluation requirements.

2.1.7 Visualization

ID	Requirement	Priority
FR-30	The system shall record all aircraft states and weapon events and export them in ACMI format for 3D replay in Tacview.	Must
FR-31	The system shall stream simulations in real time to FlightGear with a UCAV-like 3D model.	Could

Table 8. Visualization requirements.

2.2 Non-Functional Requirements

ID	Requirement	Priority
NFR-1	**Performance (inference):** a single policy decision shall take less than 10 ms on a single core of a standard desktop CPU.	Must
NFR-2	**Performance (simulation):** a single headless environment instance shall run at least 10 times faster than real time.	Must
NFR-3	**Timing:** the flight dynamics shall be integrated at a rate of at least 60 Hz, and low-level policies shall act at a rate of approximately 10 Hz.	Must
NFR-4	**Training infrastructure:** training shall run multiple environment instances in parallel on available CPU cores, save checkpoints periodically, resume interrupted runs from the latest checkpoint and log training metrics (episode reward, losses, win rates).	Must
NFR-5	**Reproducibility:** given the same configuration, code version and random seed, an evaluation run shall produce identical results.	Must
NFR-6	**Reliability:** geometry, weapon and reward computations shall be covered by unit tests.	Must
NFR-7	**Maintainability:** the code shall be modular, version-controlled with Git and documented, with each module's purpose and interface described.	Must
NFR-8	**Portability:** the system shall run with Python 3.10 or later and shall not require a GPU for inference.	Should
NFR-9	**Usability:** training, evaluation and replay export shall each be started with a single command using a configuration file.	Should
NFR-10	**Licensing:** all reused third-party components shall be used in accordance with their licenses and cited.	Must

Table 9. Non-functional requirements.

3. References

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